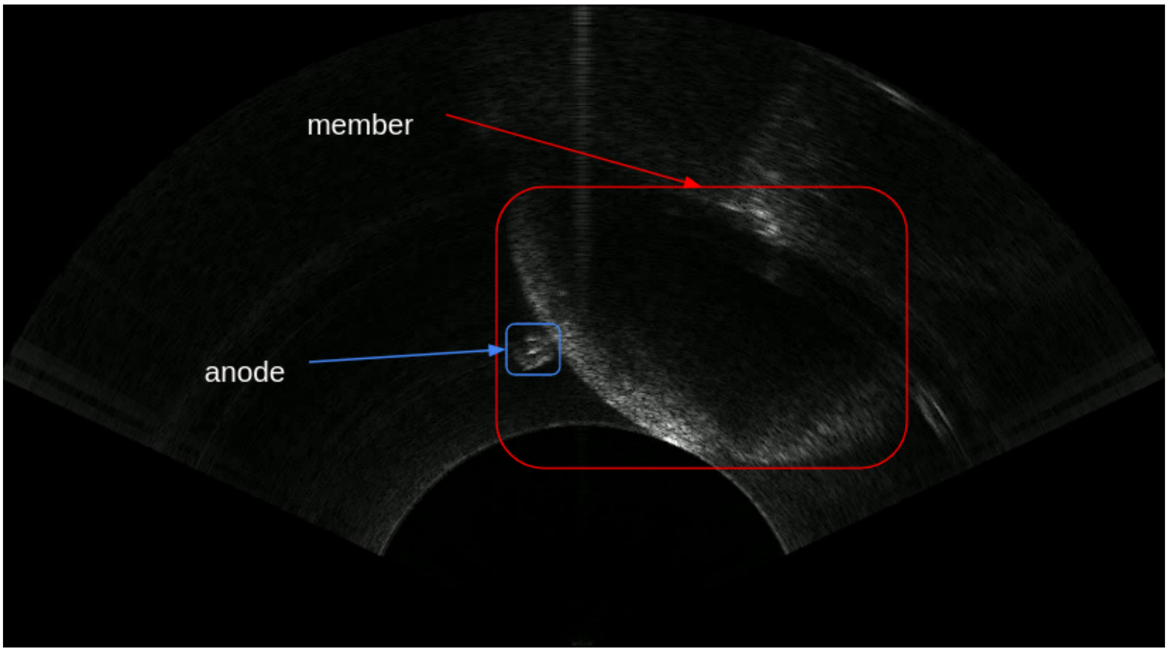
Task 2: Machine learning for image segmentation

A sonar image dataset (annotated) in the folder interview_dataset ML. The data format provided is COCO 1.1, in the location 'annotations/instances_default.json'. There are 2 classes in the data, member and anode, as shown below.



Member with Anode (Sonar)

You are given a labeled image dataset and a predefined semantic segmentation model. Your task is to train the model to accurately segment **two target classes member and anode** while treating all other object types as background. Helper functions for training and a basic loss function are provided. You must **not change the model type** but you may modify the training process. A hidden test set will be used to evaluate the final performance of your trained model. In your submission, please include the following:

- A brief explanation discussing your insights about the dataset
- Any challenges encountered and the changes you made to improve training performance.
- Your complete training code,
- Loss Curves
- Final IoU and F1 Scores.

The IoU and F1 scores for the *member* and *anode* classes should be reported separately. The evaluation will be based on segmentation accuracy on the hidden test set, correctness of implementation, clarity of analysis and overall reproducibility of your results.

Disclaimer: The aim of this task is to evaluate your fundamentals of ML/DL, rather than the actual convolutional model that you could have trained. Hence, we advise not to waste too much time and electricity on excessive model training. Moreover, we will not accept achieving double descent as a valid observation for this task.